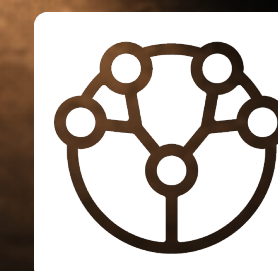




The Forgotten Users of Recommender Systems

Methodology First - Rethinking Research
Assessment in RecSys (FRAME) Workshop

Sam Pieters, Basil Rommens, Noah Daniëls, and Bart Goethals
28 september 2026

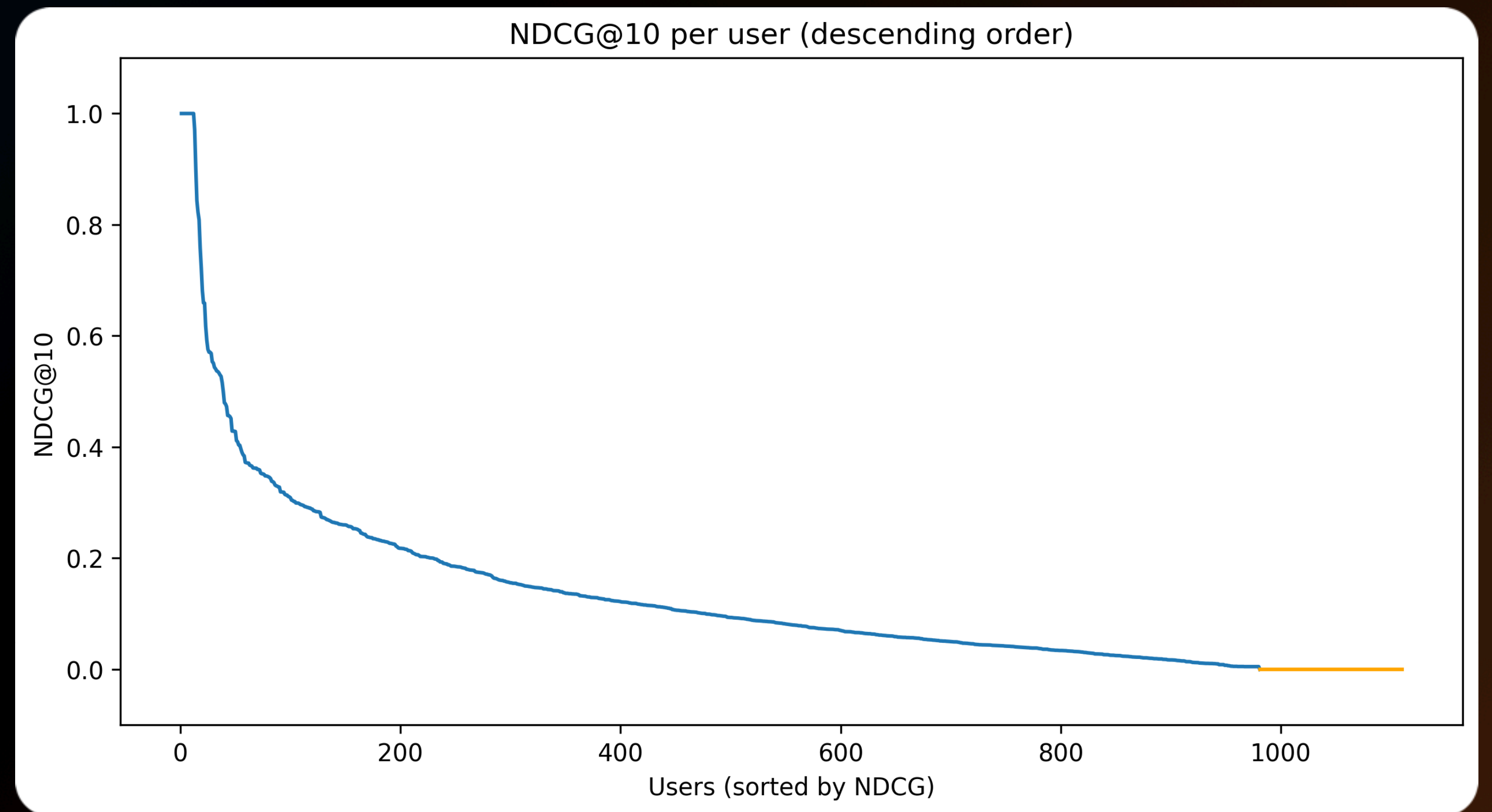


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| Adrem | Adrem Data Lab

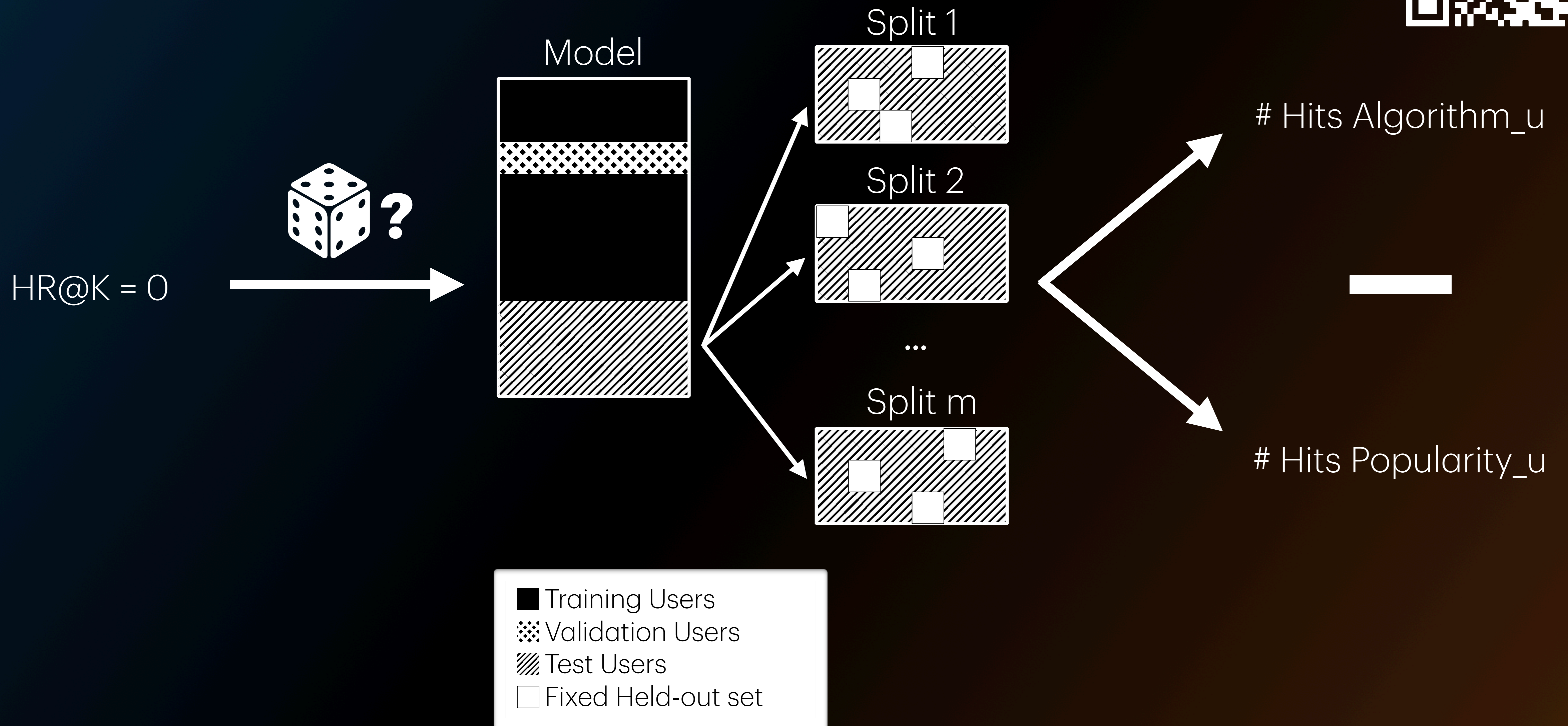
Researchers often overlook users



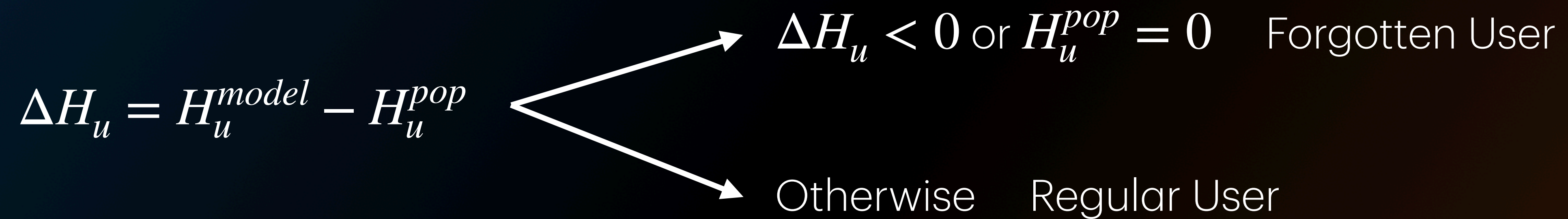
	NDCG@10
Algorithm A	0.0826
Algorithm B	0.0797
Algorithm C	0.0790
Algorithm D	0.0716
Algorithm E	0.0451
Algorithm F	0.0843
Algorithm G	0.1147
Algorithm H	0.1082
Algorithm I	0.1027



When is a user forgotten?



When is a user forgotten?

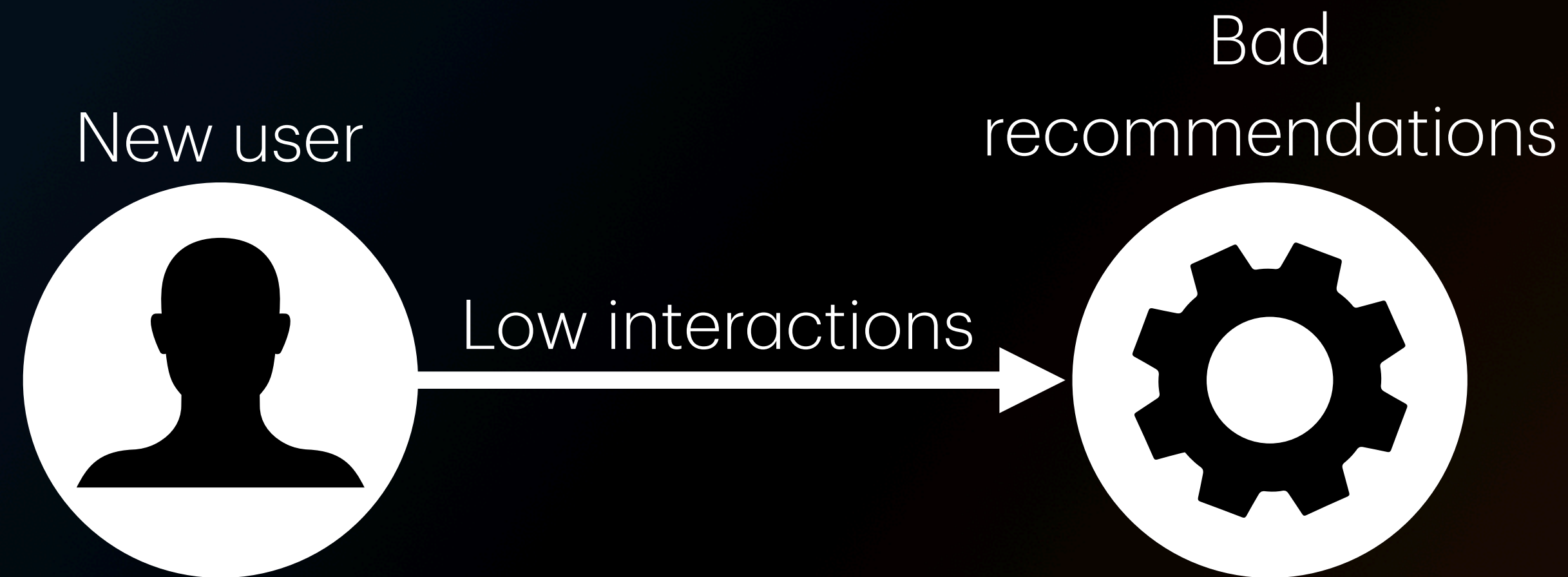


Forgotten User = "A user who receives

- **fewer hits from a specific recommender than from a popularity recommender.**
- Or **no hits from either algorithms**

across **multiple test splits.**"

Forgotten users $\neq$ Cold-start users



- **Movielens-1M:** < 20
- **Amazon PrimePantry:** < 5
- **CiteULike:** < 10
- **Netflix:** < 20

Combining algorithms



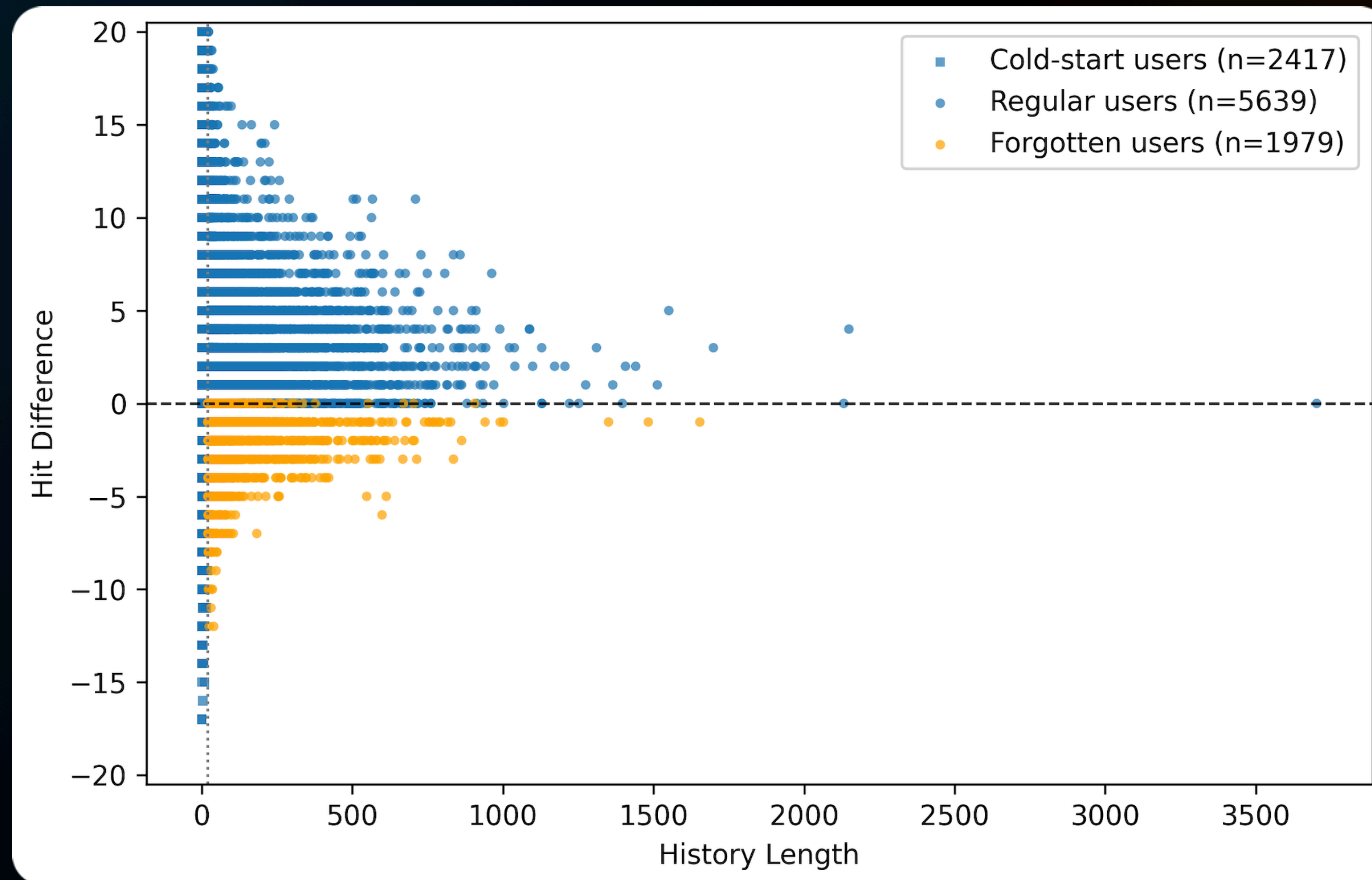
	Algorithm A	Algorithm B	Algorithm C	Greedy
Split 1	0	1	1	1
Split 2	1	0	1	1
Split 3	0	0	0	0
...	...	...	...	...
Split m	1	0	0	1

Research Questions



- **RQ1:** How prevalent are forgotten users across different recommendation datasets?
- **RQ2:** To what extent do different recommendation algorithms forget the same users?
- **RQ3:** Can combining multiple recommendation algorithms eliminate forgotten users?

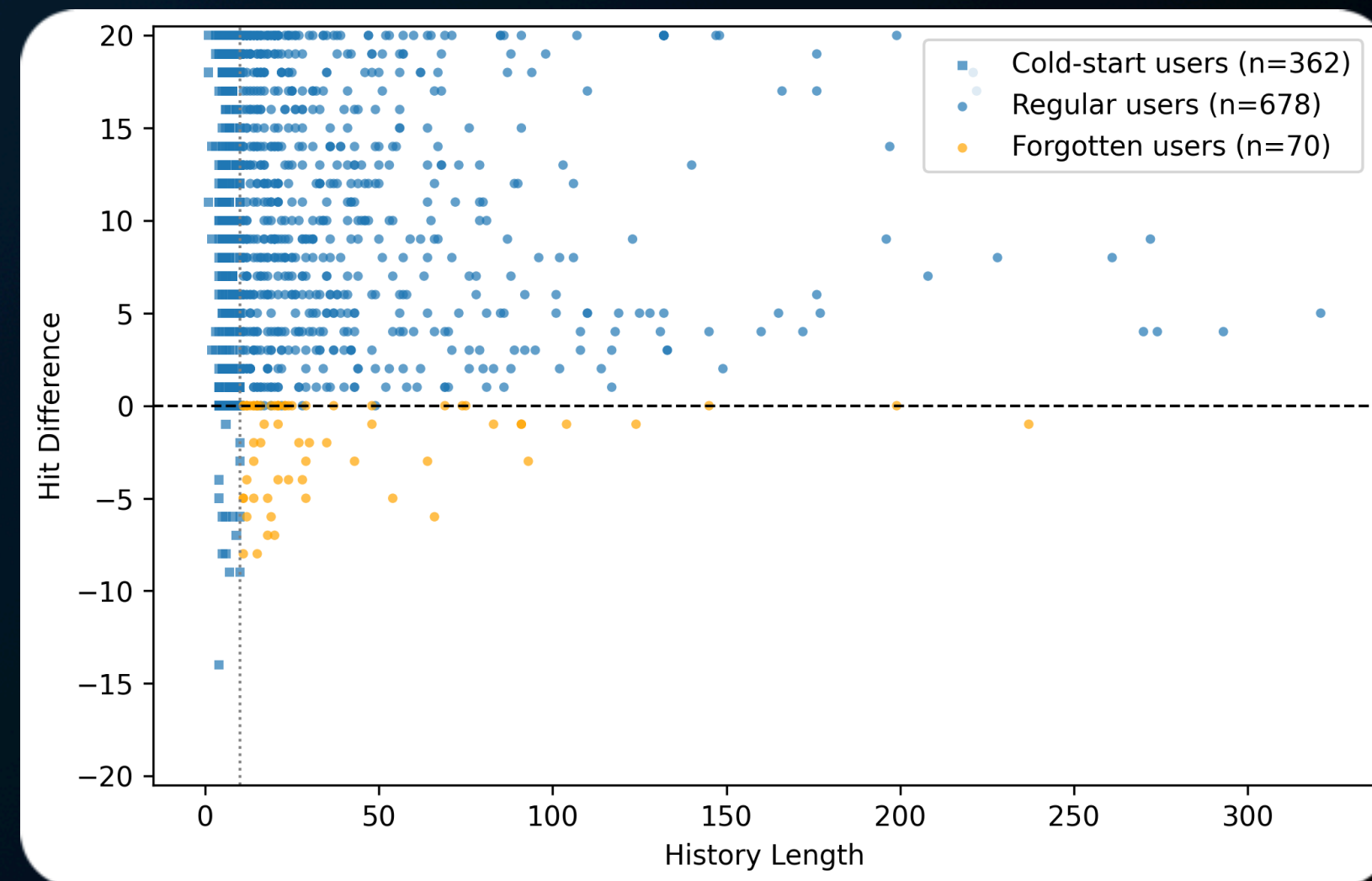
Forgotten users vary in history length



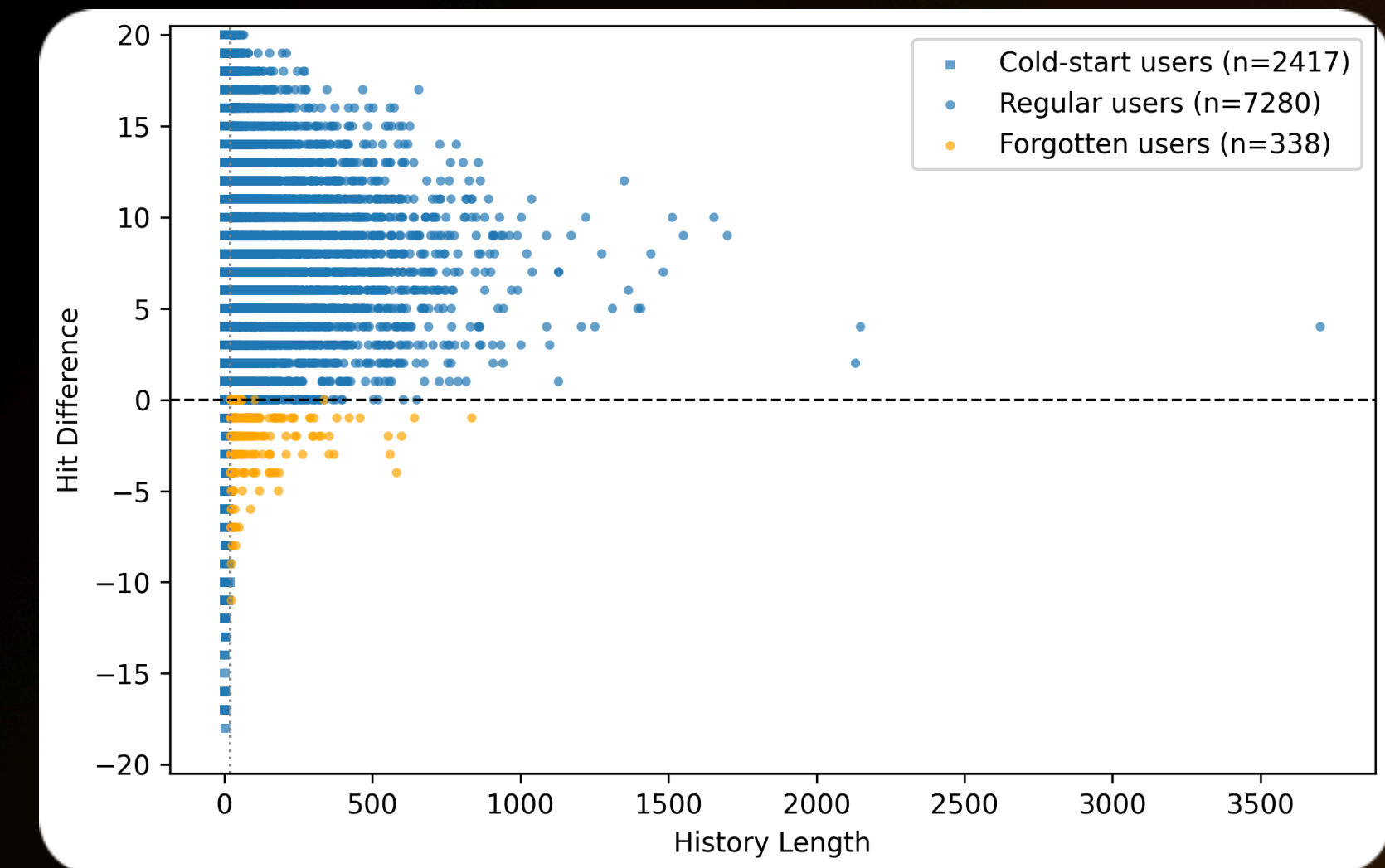
Examples of Forgotten Users



ItemKNN for CiteULike



EASE for Netflix



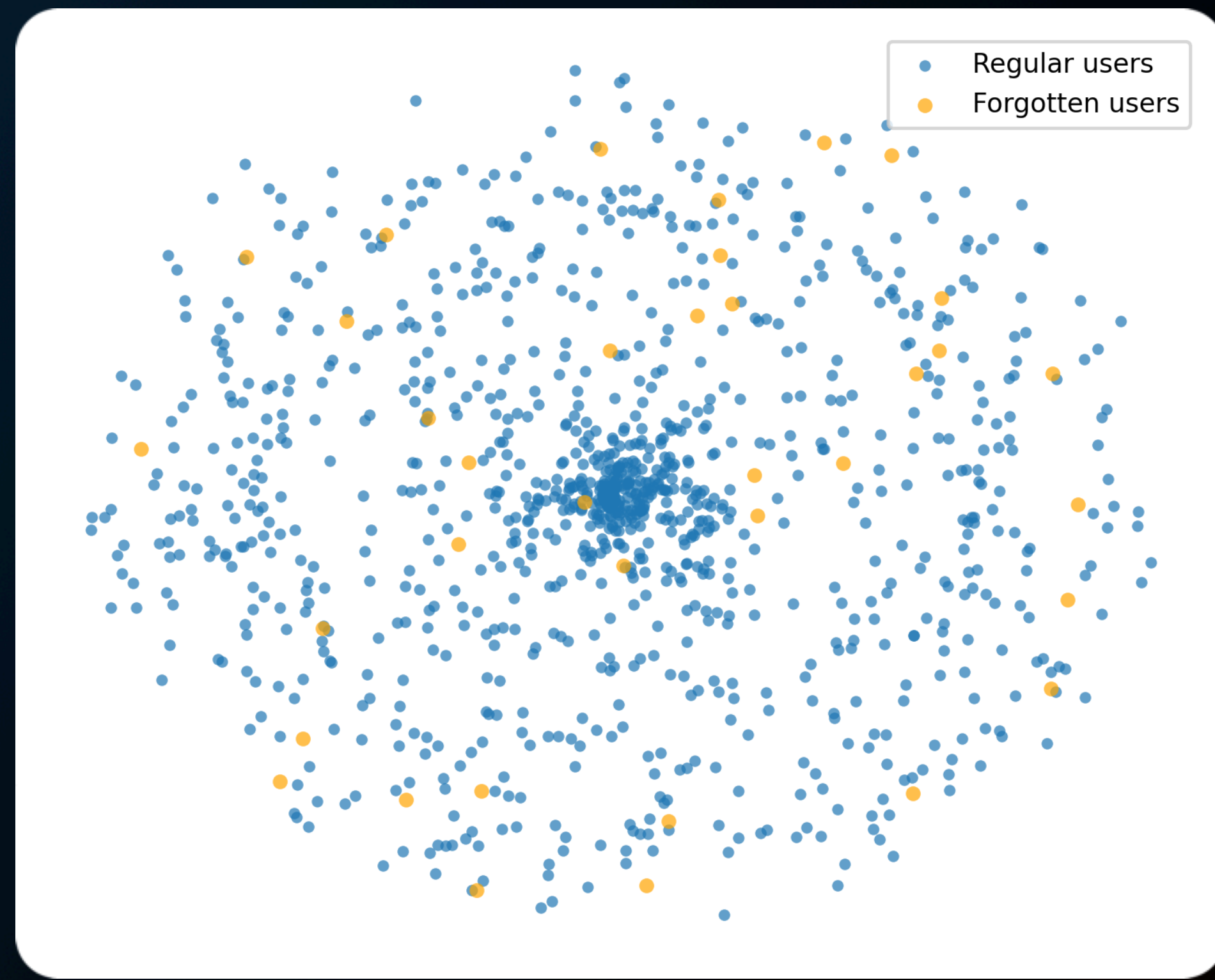
Percentage of forgotten users

Dataset	ItemKNN	EASE	SVD	Greedy
ML-1M	13.66	3.56	4.64	0.58
Amazon PrimePantry	53.58	38.91	41.00	17.32
CiteULike	6.30	3.87	6.57	1.35
Netflix	19.72	3.37	5.60	1.37

No simple pattern

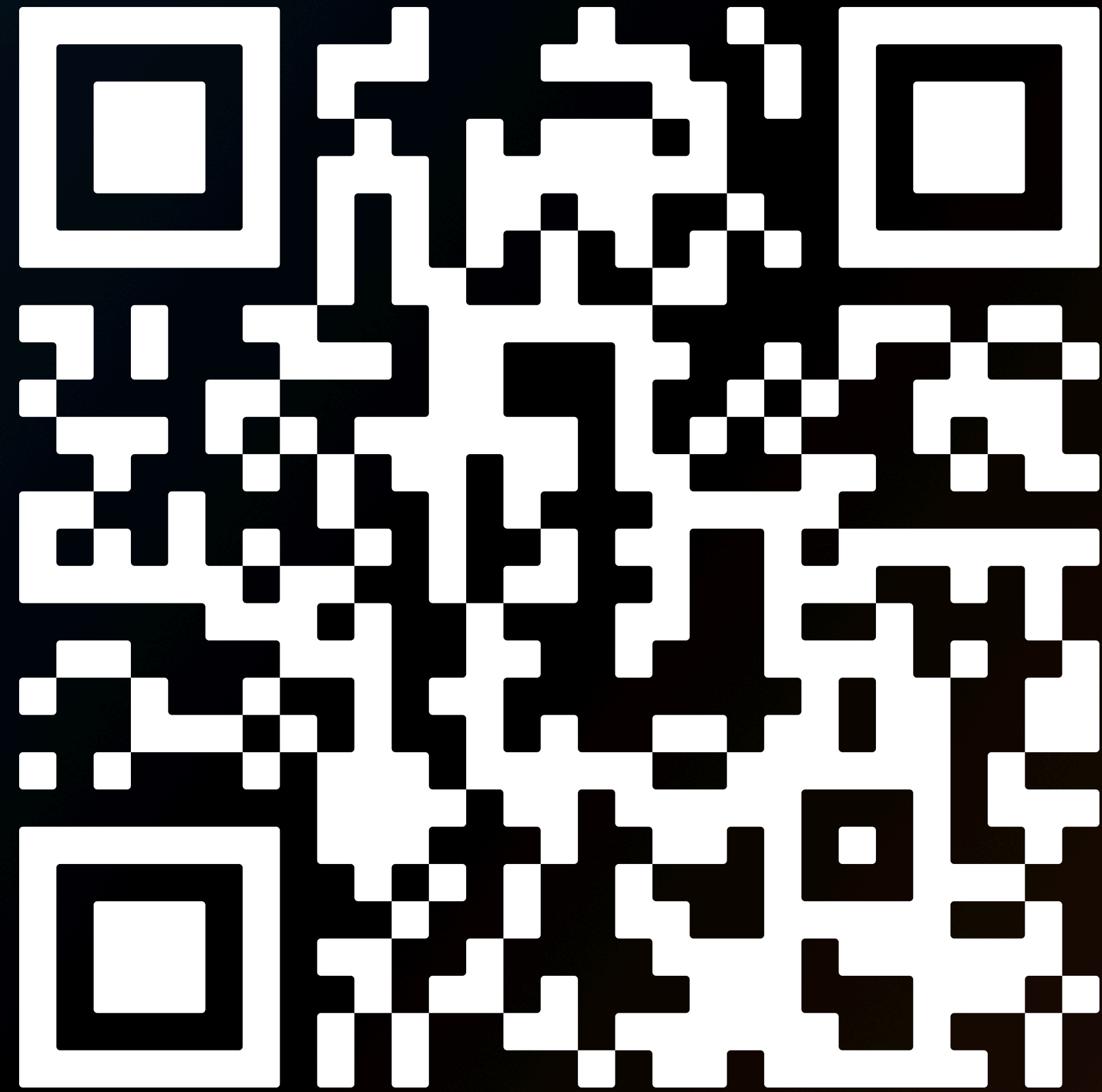


ItemKNN for CiteULike



EASE for Netflix





Paper, Code & Presentation